

Detecting and Mitigating Bias in Home Loan Approvals Using Machine Learning

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<https://github.com/dameyer/mads-w-k-d>

December 27, 2025

1 Introduction

Our goal for this paper is to develop a machine learning model that predicts home loan approvals while detecting and mitigating bias, specifically racial bias. Using the Home Mortgage Disclosure Act (HMDA) dataset, we aim to build a fair and transparent model that can improve on existing loan approval systems used by financial institutions.

Most lenders use automated underwriting systems like Desktop Underwriter (DU) or Loan Product Advisor (LPA) to decide approvals based on historical data. While these models are effective, they are often black-box systems, making it difficult for borrowers to understand decisions. Additionally, because they are trained on past loan approvals, they can reinforce historical biases against certain demographic groups.

We propose a more transparent and interpretable machine learning model that not only predicts loan approvals but also identifies and mitigates bias in the decision-making process. By incorporating bias detection techniques (e.g., disparate impact analysis) and bias mitigation strategies (e.g., reweighing, adjusting decision thresholds), we aim to ensure fairer outcomes while maintaining accuracy. We hope that this project will add meaningful contributions to achieving fairness in machine learning models. As well as make a difference in the fairness of the home loan approval process.

2 Related Work

Recent research highlights significant racial and ethnic disparities in home loan approvals, even after accounting for financial qualifications. Srivastava [Sri21] analyzed HMDA data and found that Black and Hispanic applicants were more likely to be denied loans than White applicants, even when controlling for key financial variables.

A report by The Markup [The21] also examined mortgage approval outcomes using the HMDA dataset and concluded that major lenders' algorithms disproportionately denied applicants of color. Despite controlling for factors like income, debt, and loan size, disparities persisted. The report criticized the black-box nature of these automated underwriting systems, which offer little transparency into how decisions are made.

Pavón Pérez [PP23] emphasized the necessity of fairness-aware and interpretable machine learning models in financial decision-making. Their research advocated for the use of bias mitigation techniques such as reweighing and group-specific threshold adjustments to reduce discriminatory impacts in lending practices.

Our project builds on these foundational studies by developing a transparent and interpretable machine learning model using the HMDA dataset. We selected logistic regression and random forest models due to their balance of interpretability, performance, and broad adoption. These models allow us to diagnose disparities in prediction outcomes and serve as a benchmark for evaluating the fairness of future bias-mitigation strategies.

3 Methods

3.1 Dataset

Our methodology focuses on building a fair and interpretable machine learning model using the Home Mortgage Disclosure Act (HMDA) dataset. The HMDA dataset is quite robust with just one year having over 20 million individual loan application records. To account for this, we filtered the dataset to include data from Wake County, North Carolina—selected for its racial diversity and manageable size. The ultimate goal is to detect and mitigate racial bias in home loan approval decisions.

3.2 Model Selection

For this project, we utilized two machine learning models: Logistic Regression and Random Forests.

We decided to implement a logistic regression model for its baseline performance as well as its efficiency. Logistic regression is a very strong model to use for classification problems. While it took some work, we used label encoding, ordinal encoding, and one hot encoding to transform the data in a way that logistic regression could use. We were able to get a baseline accuracy of 96.65 percent for the logistic regression model.

We also used random forest model for our project. We used random forest for its high accuracy and its ability to highlight which variables are the most influential in decision making, adding interpretability.

With having the added benefit of being able to rate the most important features, we needed to do some analysis to figure out what features would be the best to use. We learned that interest rate was the most important feature. It seems strange that interest rate influences loan approval, since lenders set it. However, the model identified strong patterns linking high interest rates with higher denial rates. We were also able to identify that income, loan amount, debt-to-income ratio, and loan purpose were also some of the highest rated important features.

We were able to achieve an accuracy of 96.99 percent with the random forest model. For that reason we chose to focus on mitigating bias in the random forest model instead of the logistic regression model.

3.3 Data Preprocessing

We began with a raw dataset named `HDMA`, which contained over 99 columns and 76,111 entries. The dataset included many irrelevant or severely incomplete variables that could introduce noise or bias into the modeling process. Therefore, our first step involved extensive data cleaning and column selection.

3.3.1 Missing Data Handling

We calculated the percentage of missing values across all columns and dropped variables with more than 95% missing data, including `denial_reason-4`, `aus-4`, and others. For features with moderate missingness (between 30% and 50%) such as `interest_rate`, `loan_to_value_ratio`, and `debt_to_income_ratio`, we applied median imputation to ensure robustness against outliers. Categorical variables such as `conforming_loan_limit` and `census_tract` were imputed using the mode. For sensitive fields like `co-applicant_age_above_62`, we imputed the value “Unknown” to preserve the presence of missingness without introducing potential bias.

3.3.2 Data Type Conversion

Several important features such as `interest_rate` and `rate_spread` were read as object types due to inconsistencies in the raw file. We used `pandas.to_numeric()` to convert these fields to numeric types, coercing invalid entries to NaN before applying imputation.

3.3.3 Feature Selection

After cleaning, we retained 31 relevant columns in a filtered version of the dataset named *HDMA_filtered*. This dataset included borrower demographics, financial attributes, loan characteristics, and decision-related variables. These columns were chosen specifically to support downstream fairness analysis and predictive modeling.

3.3.4 Outlier Removal

For bias visualization, we created a separate working dataset called *feature_disparities_df*. To reduce distortion, we excluded entries with extreme or unrealistic values—such as incomes above \$500,000 or loan-to-value (LTV) ratios above 150.

3.4 Exploratory Bias Analysis

We explored disparities across racial groups by visualizing key financial features using both box plots and violin plots. The variables examined included income, debt-to-income ratio (DTI), loan amount, interest rate, and loan-to-value ratio (LTV). These plots revealed broad consistency in patterns across races but also highlighted certain disparities. For instance, violin plots showed tight clustering around key regulatory thresholds for DTI and interest rates, while box plots made income and loan amount variability across groups more visible.

We paid particular attention to preventing misleading visual trends caused by outliers. Without proper filtering, extreme values in loan amount and LTV would have hidden the spread of typical observations. By capping visualizations at realistic bounds (e.g., $LTV \leq 150$), we ensured a clearer comparison of distributional patterns across demographic groups.

3.5 Modeling Approach

We implemented two baseline classification models: logistic regression and random forest. The target variable was `action_taken`, which captures whether a loan was approved or denied. Predictor variables included borrower income, DTI, loan amount, interest rate, LTV, and encoded demographic features such as `derived_race` and `derived_ethnicity`.

We selected logistic regression for its simplicity and interpretability, which allows us to clearly examine how each feature—such as income, debt-to-income ratio, or race—impacts loan approval outcomes. In contrast, we chose random forest for its higher accuracy and ability to model more complex, non-linear relationships in the data. These two models provide a strong baseline for understanding performance and identifying any disparities in predictions across demographic groups. They will also guide the next phase of our work, where we plan to apply fairness-aware techniques such as reweighing and group-specific threshold adjustments to mitigate potential bias.

4 Preliminary Results

4.1 Model Accuracy and Evaluation

After completing data cleaning and preprocessing on the HMDA dataset, we trained two classification models—logistic regression and random forest—to predict whether a home loan would be approved. The target variable, `action_taken`, was recoded to represent binary outcomes: one for approved loans and zero for denied loans. The training and test sets were carefully stratified to maintain class distribution. Preliminary results from both models indicate high performance in terms of accuracy and predictive power, while exploratory bias analysis has revealed areas of potential concern regarding fairness.

The logistic regression model achieved an overall accuracy of 96.65%, demonstrating strong baseline performance. Its simplicity and transparency make it a useful benchmark model for assessing fairness. In parallel, the random forest classifier yielded slightly higher accuracy at 96.99%, using 100 estimators and default hyperparameters. The random forest model was evaluated on a test set containing 10,936 records, and its predictions were further assessed using precision, recall, and F1-scores for both approved and denied classes.

For denied loans, the model achieved a precision of 0.89, meaning that when the model predicted a denial, it was correct 89% of the time. The recall for this class was 0.95, showing that the model successfully identified 95% of all actual denials. This resulted in an F1-score of 0.92, indicating a strong balance between precision and recall. For approved loans, the model achieved a near-perfect precision of 0.99 and a recall of 0.97, with an F1-score of 0.98. These metrics confirm that the model is both accurate and well-calibrated across decision classes.

4.2 Confusion Matrix Insights

An analysis of the confusion matrix provides further insight into model performance. Of the 10,936 test records, the model correctly predicted 1,825 denials and 8,782 approvals. It mistakenly approved 97 loans that should have been denied, and mistakenly denied 232 loans that should have been approved. These errors are relatively small compared to the total number of predictions and suggest that the model generalizes well to unseen data.

4.3 Feature Importance and Interpretability

In terms of feature importance, the most influential predictor in the random forest model was the interest rate. This result initially seemed counterintuitive, given that lenders set the interest rate rather than the applicant. However, further inspection revealed that higher interest rates were often correlated with denial decisions, likely because they serve as indirect indicators of borrower risk. For example, loans with excessively high interest rates may exceed regulatory cost thresholds or become unaffordable relative to the borrower’s debt-to-income ratio, both of which can lead to denials. In the absence of direct credit scores, the model appears to treat interest rate as a proxy for borrower creditworthiness.

Other important features included income, loan amount, and debt-to-income ratio. Higher income was associated with a greater likelihood of approval, while larger loan amounts were associated with increased risk and thus lower approval rates. Debt-to-income ratio, a well-known metric of financial stress, also played a significant role in the model’s predictions. Loan purpose was another influential factor, with loans intended for home purchase more likely to be approved than those for refinancing or other purposes.

4.4 Raw Denial Counts and Real-World Impact

These findings were corroborated by raw denial counts. Although White applicants comprised the largest share of the dataset with over 29,000 records, they had a denial count of 4,047—approximately one in seven. In contrast, Black applicants had 6,584 records and 1,995 denials, translating to nearly one in three. Even small groups like Native Hawaiian or Other Pacific Islander, with only 100 records, experienced 35 denials. When comparing raw counts and proportional rates across race, ethnicity, and sex, we found that systemic disparities persist, especially for Black, Hispanic, female, and Indigenous applicants. Joint applicants, on the other hand, consistently experienced lower denial rates across all demographics, likely due to combined incomes and stronger overall financial profiles.

4.5 Visual Analysis of Financial Disparities

In our visual feature analysis, we used violin and box plots to compare distributions of income, debt-to-income ratio, loan amount, interest rate, and loan-to-value ratio across racial groups. While some distributions were similar, notable differences emerged. Interest rate and debt-to-income ratios were tightly clustered around regulatory thresholds across most groups, while income and loan amount distributions showed greater disparity. For example, Black and Hispanic applicants generally appeared to have lower income and higher debt-to-income ratios than White applicants, which may partially explain the approval gap. To ensure clearer visualization, we removed extreme outliers such as loan-to-value ratios exceeding 150 or income levels above \$500,000. This allowed for a more accurate representation of typical applicant profiles and exposed patterns that might have otherwise been masked by skewed data.

4.6 Conclusion

Together, these results highlight both the technical performance of our models and the real-world disparities present in the dataset. While our classifiers achieve high accuracy, the demographic breakdowns suggest that fairness concerns remain. These findings underscore the need for bias mitigation techniques, which will be explored in the next phase of the project.

5 Disparity Analysis by Demographic Group

Beyond model evaluation, we conducted a disparity analysis to investigate whether different demographic groups experienced unequal approval outcomes. The denial rate by race revealed significant disparities: applicants who identified as Native Hawaiian or Other Pacific Islander faced a denial rate of 35.00%, followed closely by individuals of two or more minority races at 34.23%, American Indian or Alaska Native at 33.70%, and Black or African American at 30.30%. In contrast, White applicants experienced a denial rate of only 13.81%, and joint applications (likely reflecting dual-income households) had the lowest rate at 11.65%. This suggests that racially minoritized groups are disproportionately denied home loans, even after controlling for observable financial features.

A similar pattern was observed when disaggregating results by ethnicity. Hispanic or Latino applicants faced a denial rate of 26.57%, while Non-Hispanic applicants had a much lower rate of 16.38%. Applicants whose ethnicity was not reported had a denial rate of 20.81%, while joint applications again experienced a lower rate of 13.87%. When examined by sex, female applicants had a higher denial rate (22.01%) than male applicants (19.68%), while joint applications were again associated with lower denial rates (11.99%).

6 Bias Mitigation techniques

6.0.1 Data Disparity

The subset of data that we are using from the HDMA dataset is all the bank data from Wake County, North Carolina. We chose this area because we wanted an area that was very diverse. In Wake County, 40.6 percent of the population identifies as Black, Latin, Asian, Native American, multiracial, or Other. We believed training our model on data like this would help ensure that we had enough data on each racial group to build a model that would potentially have racial bias that we could work to mitigate. As time has gone on, we are wondering if maybe choosing data from a very diverse area like Wake County has helped mitigate bias in the model. Since it is a very diverse area it may be overlooking bias in other parts of the country that are predominantly one race. We assume in areas that are predominantly one race, there may be more bias in accepting individuals for loans. We still chose to continue with this dataset however it could be insightful to use a different, less racially diverse area in the future.

To fix this, we would like to compare the data ratios we have in Wake County to the national average. We think this would be insightful to see how representative our data is and see if choosing Wake County is introducing bias into our model. We understand that there is no perfect place to pick and that we don't have the computing power to handle the national HDMA data, but we would like to see how our data compares.

This may be something that we may need to focus on after this project is completed given the time constraints that we have and our desire to focus on other bias mitigation techniques.

6.0.2 Readjusting weights

We would also like to spend some time readjusting the weights in our model to see if we can fine-tune the model to help it be more accurate. Right now we have some baseline predictions from our model showing us the percentage of probability that each race has of being approved for a loan. We would like to compare those percentages to a model with fine-tuned weights to see if that creates a higher percentage of acceptance for certain races and also to see if all races have similar approval rates.

We believe that this would be a very important mitigation technique to focus on because it will directly address the data imbalance that we may have in our Wake County, North Carolina data sample as well as adjust for historical bias that may be a part of the dataset we have.

6.0.3 Threshold change

Another idea that we had on how to mitigate racial bias in home loan approval data was to change the thresholds on our model. Looking at our preliminary results, there were different probabilities of each racial group being approved for a loan. These different groups could be getting different probabilities

due to biased data or model behavior. With us wanting to adjust the decision threshold per group, we could better balance the outcomes.

Another great benefit of exploring threshold change is that this is a postprocessing technique. That means that it will not require us to retrain the model when we are working on this step. We are using a relatively small dataset but if in the future if we were to run this on state, multiple state, or national dataset it would be a great benefit and less expensive.

7 Findings

7.1 Readjusting weights

One of the mitigation techniques that we wanted to use was readjusting weights or reweighing our model. Reweighting is a preprocessing technique that adjusts the importance (weights) of training samples based on group membership and class label. It helps reduce historical bias by giving more weight to underrepresented or disadvantaged groups, without altering feature values or labels. We specifically chose to focus on reweighting the random forest model since it had better accuracy than the logistic regression model.

In our project to accomplish this we chose to use AI Fairness 360 (aif360). Aif360 is an open-source python toolkit developed by IBM to help detect and mitigate bias in machine learning models and datasets. It is very well known and respected, so we thought it would be a great option to use in our project. In our project, we focused our reweighing process specifically on reducing bias against Black or African American applicants. Other historically disadvantaged groups (such as American Indian, Alaska Native, Native Hawaiian or Other Pacific Islander) were not included in this round of fairness correction. This targeted approach was chosen to isolate and evaluate the impact of reweighing on a single high-priority group. Future iterations could extend fairness protection to additional groups.

After reweighing our random forest model, our findings were interesting but insightful. In our original random forest model, our accuracy was 96.99 percent and it had a loan approval rate of 81.19 percent. After reweighing the model, the accuracy improved to 97.04 percent and the loan approval rate dropped to 80.85 percent. While there was a slight accuracy improvement, there was also a slight loan approval rate drop. Reweighting successfully improves fairness for the primary unprivileged group (Black or African American) without sacrificing model performance. While some disparities remain for other groups, this analysis shows a strong foundation for building responsible and equitable ML models in the financial domain.

7.2 Threshold Optimization

To further explore fairness in our loan approval model, we applied a postprocessing technique using Fairlearn's ThresholdOptimizer. This method adjusts decision thresholds for each group (e.g., racial group) to satisfy fairness constraints — in this case, Equalized Odds.

Unlike reweighing (which operates before training), ThresholdOptimizer works after the model has been trained. It does not change the model itself or its probabilities — instead, it applies group-specific thresholds during prediction to enforce fairness constraints. In this analysis we applied a fairness constraint and a sensitive attribute. The fairness constraint required equal true positive rates and false positive rates across groups and the sensitive attribute focused on the derived race.

For this, we chose to compare three models. The first model was the Original RF model with a default threshold of 0.5. Second, we used a custom threshold model using a manually chosen threshold of 0.4. Lastly, we used a Fairlearn ThresholdOptimizer applying group-specific thresholds to achieve equalized odds.

To evaluate the trade-offs between accuracy, fairness, and approval rate, three model configurations were compared. The results can be found in the appendix under Table 1, which provides detailed performance metrics across the different threshold settings and the Fairlearn optimizer.

In the following section, we will talk about the results of the three models that we tested. Focusing on what the results tell us and what we have taken away from using this threshold optimization.

The original threshold had both high accuracy and precision. It slightly favored the majority class which in our dataset was white. There was no fairness constraint applied, so disparities across groups may still persist. With the custom threshold we do see some changes. This model slightly lowers the

decision boundary to approve more applicants. There is an increase of recall and approval rate. It however still applies the same threshold to all groups, so the fairness isn't explicitly addressed. Lastly, the Fairlearn Threshold Optimizer. This model applied different thresholds per group to satisfy the equalized odds. Approval rate decreased slightly as the model became more conservative. Recall drops, but precision increases, especially for the positive class. Fairness improved across groups (racial disparities in true/false positives are reduced). While the Fairlearn's approach slightly reduces overall accuracy, it does improve fairness. It especially improves fairness for groups that may have previously been over or under approved.

Threshold Optimizer shows that postprocessing methods can enforce fairness without retraining models, Equalized odds can be achieved by adjusting how decisions are made — even if the model predictions stay the same, and it offers a way to make real-world models fairer in deployment, especially in regulated domains like finance

Fairlearn's ThresholdOptimizer allows for fairness-aware decision-making by applying group-specific thresholds to predicted probabilities. This method achieved better fairness alignment (equalized odds) at the cost of slightly reduced accuracy and approval rates. In doing so, it helps ensure that model predictions translate into equitable real-world decisions, especially for protected groups.

7.3 Threshold Sweep Analysis: Performance vs. Fairness

To further evaluate the impact of decision thresholds on both model performance and fairness, we conducted a threshold sweep from 0.20 to 0.80. For each threshold, we computed: Accuracy - "Overall prediction correctness", Approval Rate - "Percent of applicants predicted to be approved (positive class), False Positive Rate (FPR) - Percent of incorrectly approved applications among actual denials, and Disparate Impact (DI) - Approval rate of minority group divided by approval rate of White applicants. The Disparate Impact (DI) was calculated for black or African applicants and Asian applicants. White applicants were used as the reference group for fairness comparison. For a complete view of the table, refer to Table 2 in the appendix.

Disparate Impact for Black applicants remains close to the 0.80 threshold — starting at 0.856 and falling slightly as the threshold increases. Asian applicants consistently show higher parity ($DI > 0.95$), indicating fewer fairness concerns in this group. Raising the threshold lowers FPR but reduces approval rate and worsens fairness for Black applicants.

As the threshold increased, the accuracy increased slightly (more conservative decisions), approval rates decreased, and false positive rate drops sharply (fewer incorrect approvals).

Disparate impact for Black applicants gradually declines with higher thresholds, approaching and eventually falling below the 0.80 fairness threshold. Disparate impact for Asian applicants remains stable and above 0.95. This analysis highlights how small shifts in threshold values can have significant fairness implications, even when accuracy seems stable.

Threshold tuning offers a powerful tool to balance performance and fairness in model deployment. However Higher thresholds improve conservativeness but may exacerbate disparities for historically disadvantaged groups. Lower thresholds can support fairness (higher Disparate impact) but may increase false positives. Selecting a threshold should consider not only performance metrics but also the social and legal implications of disparate outcomes.

8 Conclusion

8.1 Accomplishments

When we started this project we had the goal to develop a machine learning model that predicts home loan approval while detecting and mitigating racial bias. Throughout the semester we have worked hard to accomplish this goal and we have been able to do this successfully.

We were able to implement this fairness by reweighing our model using IBM's AI Fairness 360 toolkit and by implementing threshold optimization by using Fairlearn's ThresholdOptimizer. We evaluated the impact of these implementations by comparing accuracy, approval rate, false positive rate, and disparate impact.

We learned that reweighing slightly improves accuracy and reduces disparities for Black applicants without affecting performance. We saw that threshold optimization improved the fairness across racial

groups by reducing disparities in false positive and true positive rates. We used a postprocessing techniques both impacted performance and fairness.

8.2 Contributions

What we have done in this project contributes to the application of working to eliminate bias in machine learning models by using both reweighing and threshold optimization techniques. We were able to show that accuracy does not need to drop in order to produce a fair model.

In the introduction we talked about how most lenders use Desktop Underwriter and Loan Product Advisor that reinforce historical bias against certain demographic groups. We have shown that building a machine learning model that is both transparent and accurate while lowering bias is something that is possible. We hope that this will lead to further research as well as updates to the current systems in use.

We also have shown that you are able to build a transparent model that can be accurate as well as unbiased. It would be very insightful and helpful to individuals who are working to get approved for a home loan to know and understand why they were not approved. Knowing that their model has been created with racial unbiased in mind could be helpful for people who think that racial bias may be a factor in their loan approval being rejected.

8.3 Future Direction

Even though great progress has been made, we acknowledged that there is still room for improvement. Below are some ideas that we think could be useful to research in the future.

8.3.1 Data Disparity

We talked about this a little in 5.0.1 but data disparity is something that would be very interesting to look into in the future. As we talked about, we chose Wake County, North Carolina as our dataset because of its diversity. As we were nearing the end of our project, our team had a thoughtful discussion of if this data may have been too diverse. Initially, we thought that this would be ideal but as time went on we questioned if sampling an area with greater diversity didn't show the bias that is prevalent in other parts of the nation where the area isn't as diverse.

It would be very interesting to see how our data compares to the national average. It would also be very interesting to see if choosing an area that is predominately one race (greater than 80 percent) if that would change our results.

8.3.2 Explore Past Data

Another thing that would be very interesting to explore would be checking the HMDA data five, ten, fifteen, twenty year ago. It would be very insightful to see if there appears to be a change in bias in the data over those years.

References

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Appendix: Additional Results

Metric	Original (0.5)	Custom (0.4)	Fairlearn Optimizer
Accuracy	96.97%	96.43%	94.60%
Precision	98.82%	97.75%	99.74%
Recall	97.49%	97.93%	93.70%
Approval Rate	81.32%	82.57%	77.43%

Table 1: Performance comparison of Random Forest threshold strategies and Fairlearn optimizer

Threshold	Accuracy	Approval Rate	FPR	DI (Black)	DI (Asian)
0.20	92.84%	88.10%	36.52%	0.856	0.991
0.40	96.43%	82.57%	10.56%	0.808	0.978
0.50	96.97%	81.32%	5.46%	0.804	0.972
0.80	96.57%	79.25%	0.73%	0.792	0.955

Table 2: Performance metrics for different threshold values.

GitHub Repository

GitHub Repo: [mads-w-k-d](#)